

WEB TECHNOLOGIES

COURSE CODE:CS3005

COURSE OBJECTIVES

The objective of this course is to enable the student to:

- To understand the basics of web technology.
- To learn to design web pages using HTML, CSS, Java Script.
- To learn client and server-side web development using JavaScript libraries and frameworks.
- To understand document-oriented database for web development.
- To understand the web service concepts.

UNIT 1 - WEB SITE BASICS

9

Web Essentials: Introduction to HTML, CSS, JavaScript - Data types, Arrays, Functions, Other built-in objects. DOM - Document tree traversal and manipulations, Event handling. Introduction to AJAX - Request, Response.

UNIT 2 - CLIENT-SIDE PROGRAMMING

9

Frontend Development: React JS/ React Overview, Model-View-Controller (MVC) Architecture, Introduction to ECMAScript6 (ES6), Virtual DOM, Javascript with XML (JSX), Components - Functional Component, Exporting and Importing Components, States, Props, Hooks – useState, useEffect, React Routing, React Bootstrap.

UNIT 3 - SERVER-SIDE SCRIPTING

9

Backend Development: Introduction Node.js (LTS) – Modules, HTTP Module, File System, Node Package Manager (NPM), URL, Structured Data: MySQL, and Unstructured Data: MongoDB – Connecting, Creating, and Manipulating Databases, Express.

UNIT 4 - WEB SERVICES, API AND FSD

9

Introduction- SOAP, REST, SOAP Architecture, **REST:** Architecture, Requests and Responses, RESTful APIs, OData, Micro services **Full Stack Development (FSD):** MERN Stack.

UNIT 5 - VERSION CONTROL AND DEPLOYMENT

9

Introduction to Version Control (GIT/GITHUB), APIs Security – Designing, Creating, Testing and Validating RESTful APIs, Introduction to Postman for API testing, Public Hosting, Project Deployment and Maintenance.

TOTAL HOURS: 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

Course Outcome	Max. Knowledge Level	Course Statement
CO1	K4	Analyze and Build Dynamic Websites with HTML, CSS, Javascript, AJAX.
CO2	K4	Analyze and Build Single Page Applications using React for Front-End Application Development
CO3	K4	Analyze and Build Node.js and Express for Back-End Development with Data Storage for Structured and Unstructured Data.
CO4	K4	Analyze and Build MERN Stack for Full Stack Applications with RESTful APIs
CO5	K3	Apply Version Control using Git along with Project Deployment and Maintenance.

TEXTBOOKS

1. Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “Internet & World Wide Web How to Program”, Pearson, Fifth Edition, 2012. (Unit I & V)
2. [Brad Dayley](#), [Brendan Dayley](#), [Caleb Dayley](#), “Node.js, MongoDB and Angular Web Development: The definitive guide to using the MEAN stack to build web applications”, Pearson Education, Second Edition, 2017. (Unit II,III& IV)

REFERENCES

1. Mark Pilgrim, HTMLS: Up and Running, O'Reilly, 2010.
2. Brad Dayley, Brendan Dayley, Caleb Dayley, Learning Angular: A Hands-On Guide to Angular 2 and Angular 4, Second Edition, Addison-Wesley Professional, 2018.
3. [Evan Hahn](#), “Express in Action: Node applications with Express and its companion tools”, First Edition, Manning Publications, 2015.
4. [Ki istina Chodorow](#), Shroff, Mongoddb: The Definitive Guide- Powerful and Scalable Data Storage, Second Edition, O'Reilly, 2013.
5. [SubbuAllamaraju](#), RESTful Web Services Cookbook, First Edition, O'Reilly, 2010.